Chaos in Perspective

September 28, 2026

1. **History.** Brahe data, Kepler orbits, Newton orbits, Poincaré three-body motion.
2. **Evolution of thought.** Poincaré orbits should be simple. Poincaré oops they may not be! Lorenz and Rössler give simple chaotic systems. Chaos is everywhere
3. **Tools** Uniqueness, Sensitivity, Poincaré section, model ODE, return map, return time, calc tools, Newton's method, singular and eigenvalues.
4. **Today: stability.** Many winding numbers in Rössler computation with $c = 5.7$! Bifurcations from $1 \to 2$ loop orbits at smaller c: Computation and Stability!

"Much light may be thrown on some of these questions by consideration of stability and instability."

Maxwell, 1873